

第三章作业

3.3

又对上层节点应用KCL定律, 则有

$$8 = \frac{V_0}{10} + \frac{V_0}{20} + \frac{V_0}{30} + 20 + \frac{V_0}{60}$$

解得 $V_0 = -60V$

则可得 $I_1 = \frac{V_0}{10} = -6A$, $I_2 = \frac{V_0}{20} = -3A$, $I_3 = \frac{V_0}{30} = -2A$, $I_4 = \frac{V_0}{60} = -1A$

(此处以向下为正方向)

3.10

采用节点电压法, 设下方导线为参考地, 左节点电压为 V_1 , 中节点电压为 V_2 , 右节点电压为 V_3 , 则

对左节点: $\frac{V_3 - V_1}{1} = 4 + \frac{V_1}{8}$ ①, 对中节点: $4 = 2 \times \frac{V_1}{8} + \frac{V_2}{2}$ ②, 对右节点: $2 \times \frac{V_1}{8} = \frac{V_3 - V_1}{1} + \frac{V_3}{4}$ ③

联立①②③可得 $\begin{cases} V_1 = 32V \\ V_2 = -8V \\ V_3 = 32V \end{cases}$, 则 $I_0 = \frac{V_1}{8} = 4A$

3.30

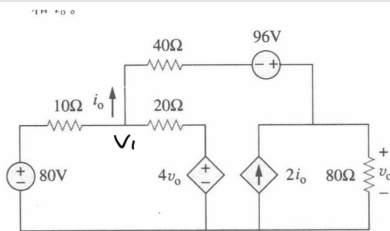


图 3-79 习题 30 图

在左侧节点, 有 $\frac{V_1 - 80}{10} + \frac{V_1 - 4V_o}{20} + \frac{V_1 - (V_o - 96)}{40} = 0$

即 $0.175V_1 - 0.225V_o = 5.6$ ①

右侧节点, 有 $\begin{cases} 2i_o + i_o = \frac{V_o}{80} \\ i_o = \frac{V_1 - (V_o - 96)}{40} \end{cases}$ 即 $-0.075V_1 + 0.0875V_o = 7.2$ ②

联立①②可得 $V_o = -1075.2V$, $V_1 = -1350.4V$

$i_o = -4.48A$

3.41

对网孔 1: $10i_1 - 6 + 2(i_1 - i_2) = 0$ 即 $12i_1 - 2i_2 = 6$

对网孔 2: $4i_2 + 2(i_2 - i_1) + (i_2 - i_3) + 8 = 0$ 即 $-2i_1 + 7i_2 - i_3 = -8$

对网孔 3: $6 + 5i_3 - 8 + (i_3 - i_2) = 0$ 即 $-i_2 + 6i_3 = 2$

解得 $\begin{cases} i_1 = 0.33A \\ i_2 = -1.0256A \\ i_3 = 0.1624A \end{cases}$ 而 $i = i_3 - i_2 = 1.188A$

3.62

又对整体回路视作超回路, 则

$$100 - 4000i_1 - 8000i_2 - 2000i_3 - 40 = 0$$

又有 $i_2 - i_1 = 0.004A$, $i_2 - i_3 = 2i_1$

可联立解得 $\begin{cases} i_1 = 2mA \\ i_2 = 6mA \\ i_3 = 2mA \end{cases}$